

DP Computer Science: A 4.3.10 / A 4.4.1 Model Selection & Ethical Implications

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the **importance of model selection** and why different algorithms perform differently depending on the problem and data
 - Describe the **"No Free Lunch" theorem** in the context of machine learning
 - **Analyse** a problem scenario and recommend a suitable machine learning model, justifying the choice based on factors like **interpretability, complexity, and data characteristics**
 - Begin to **integrate** the technical choice of a model with its potential **ethical implications**
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Key Vocabulary

Term	Definition
Model selection	The process of choosing the most appropriate machine learning algorithm for a given problem
No Free Lunch theorem	A theorem stating that no single algorithm is the best for every problem; performance depends on the problem and data
Interpretability	The ability to understand and explain how a model makes decisions
Predictive power	The accuracy or performance of a model in making predictions
Black box model	A model that is difficult to interpret or explain (e.g., deep neural networks)

Term	Definition
White box model	A model that is transparent and easy to explain (e.g., decision trees)
Bias	Systematic error in a model that can lead to unfair or discriminatory outcomes
Transparency	The degree to which a model's decision-making process is open and understandable
Accountability	The responsibility of model developers and users for the outcomes of their models

Approaches to Learning (ATL) Elements

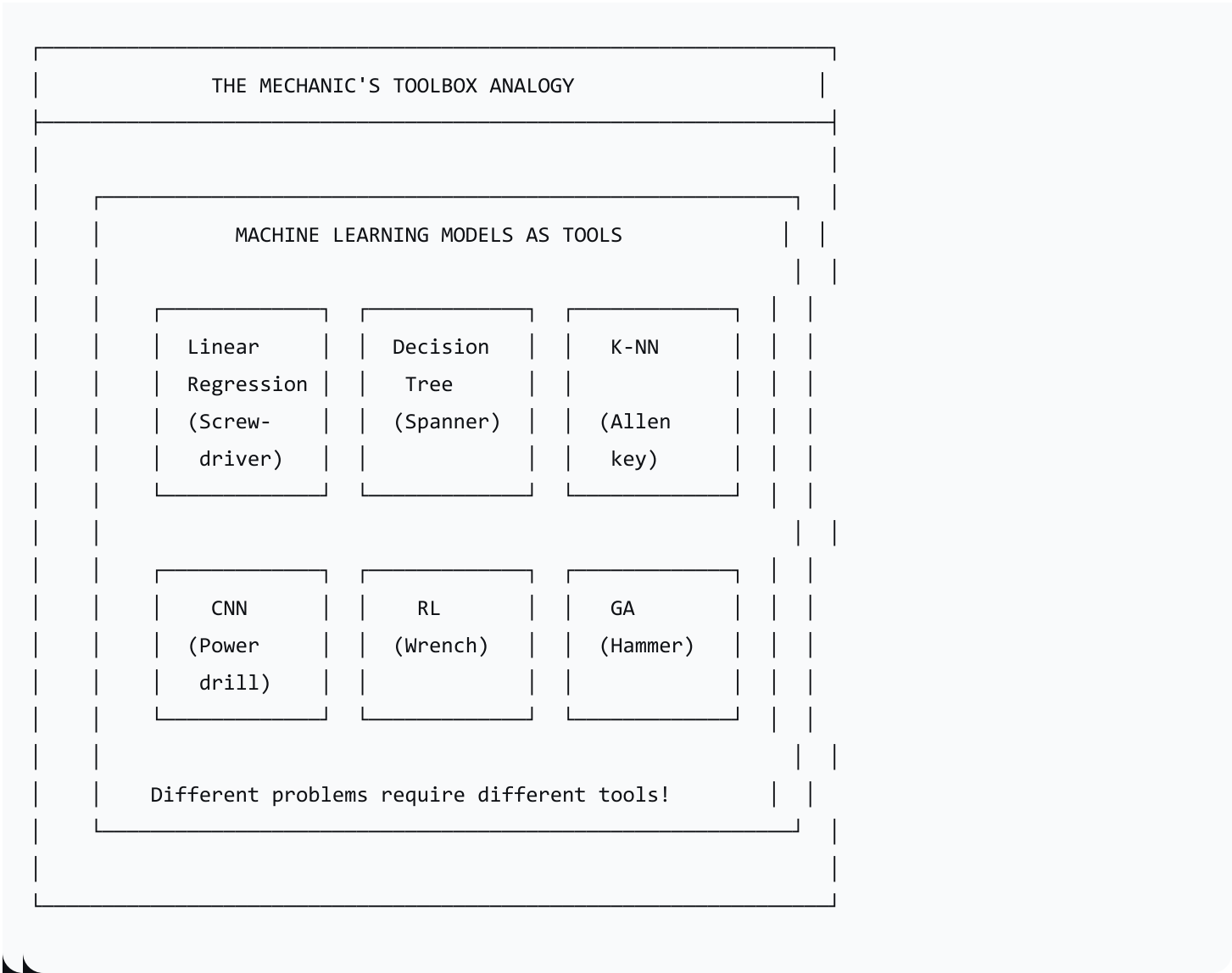
Skill	Description
Thinking (Evaluation & Critical Thinking)	Evaluating the strengths and weaknesses of different models; making reasoned judgments based on multiple criteria
Communication	Articulating a clear and concise justification for a chosen model, considering both technical and contextual factors

2. Introduction: The Mechanic's Toolbox

The Hook

"If you have a car problem, a good mechanic doesn't just grab a hammer. They diagnose the problem and choose the right tool. How do we choose the right tool in machine learning?"

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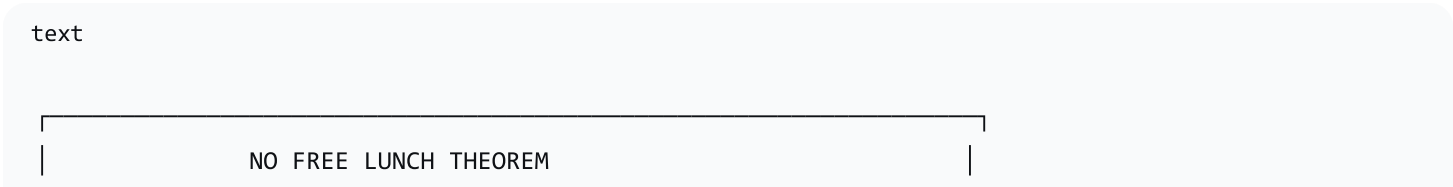


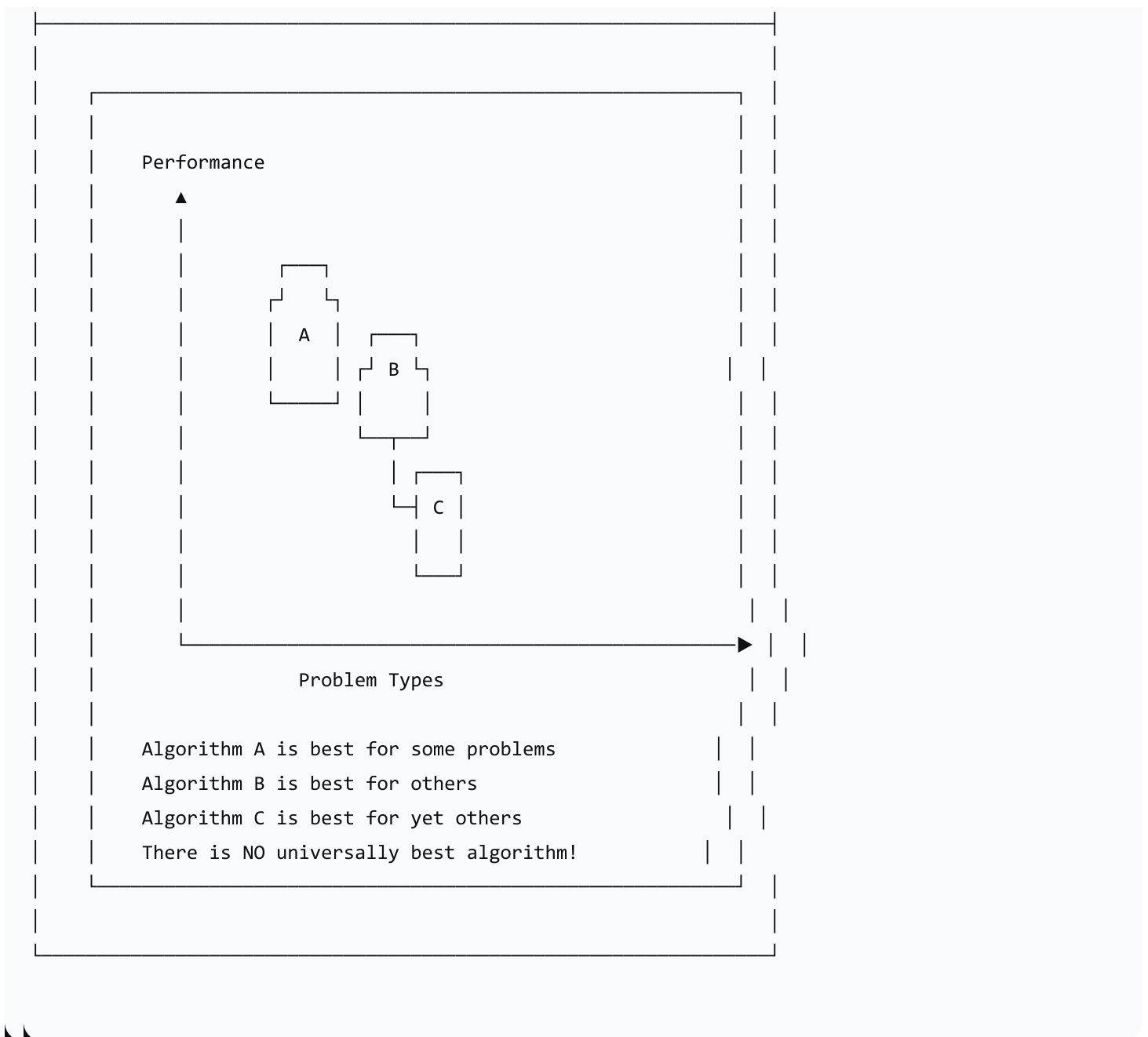
3. The No Free Lunch Theorem

What is the No Free Lunch Theorem?

The "No Free Lunch" theorem states that no single machine learning algorithm is the best for every problem. The performance of an algorithm depends on the specific problem, the data, and the desired outcome.

Visual Representation





Key Implications

Implication	Description
No Universal Best	No model works best for every problem
Context Matters	The choice depends on the problem, data, and desired outcome
Trade-offs	There is always a trade-off between different factors (accuracy vs. interpretability)

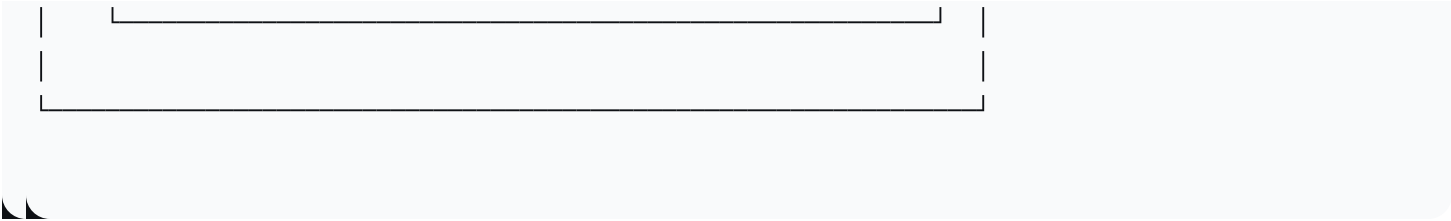
Implication	Description
Evaluation is Essential	You must evaluate models on your specific data and problem

4. Model Selection Criteria

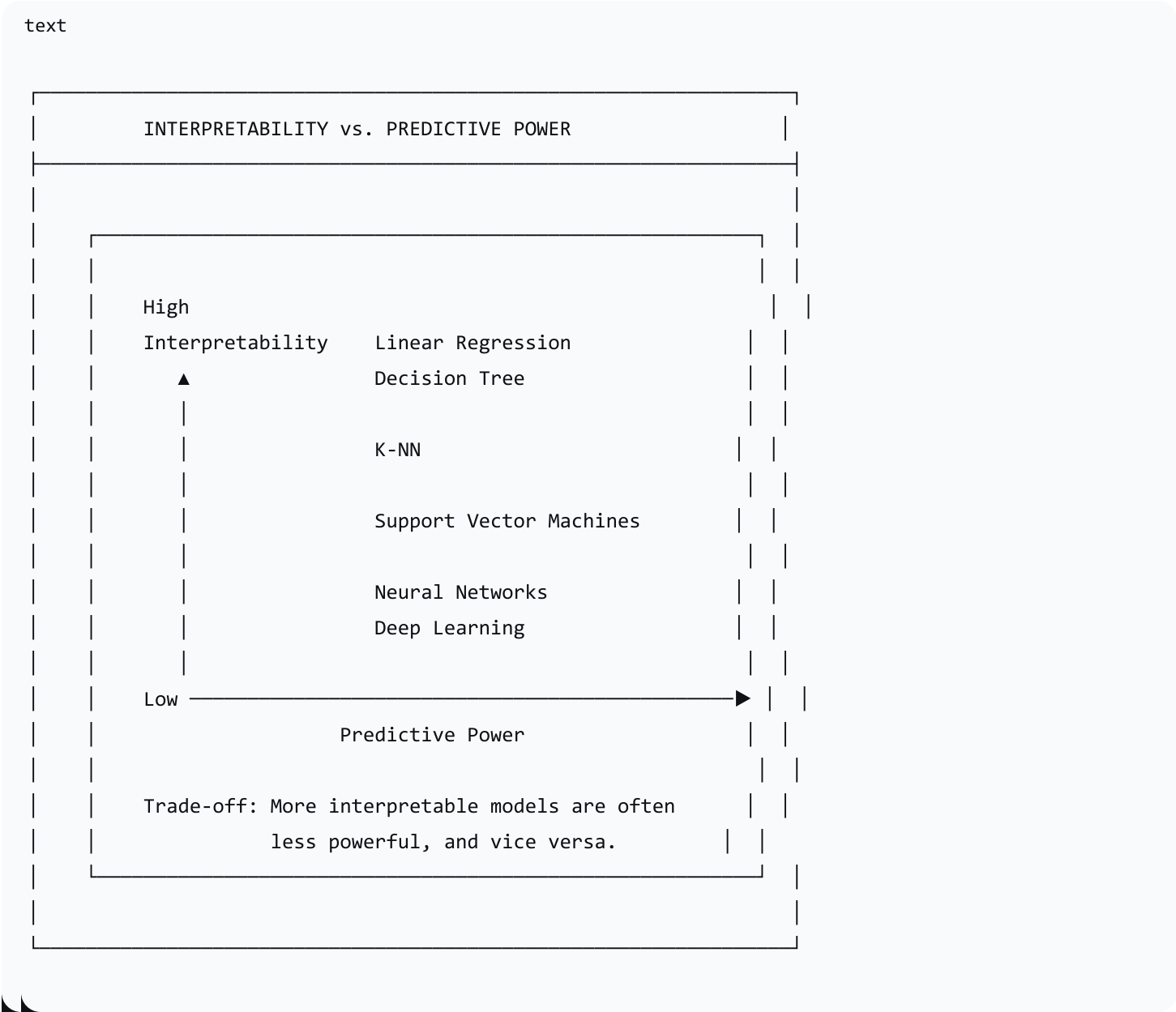
Key Factors to Consider

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MODEL SELECTION CRITERIA		
1. PROBLEM TYPE		
• Regression (predict continuous values)		
• Classification (predict categories)		
• Clustering (find groups in data)		
• Reinforcement Learning (sequential decisions)		
2. DATA CHARACTERISTICS		
• Size (small vs. large)		
• Linearity (linear vs. non-linear)		
• Number of features (low vs. high)		
• Labeled or unlabeled		
3. DESIRED OUTCOME		
• Interpretability vs. Predictive Power		
• Speed (training and inference)		
• Resource constraints (memory, compute)		
• Regulatory requirements		



Interpretability vs. Predictive Power



When to Prioritise Each

Priority	When to Choose	Example
Interpretability	When decisions must be explained (regulation, healthcare, finance)	Loan approval, medical diagnosis
Predictive Power	When accuracy is the only goal	Image recognition competition, spam detection
Balance	When both matter but neither is the only concern	Customer churn prediction

5. Activity: "Chief Data Scientist for a Day"

Instructions

1. In groups, you are the **Chief Data Scientist**
2. Analyse three distinct problem scenarios
3. For each scenario, choose the **best machine learning model** from all the models you have learned
4. Write a short **justification** based on:
 - Problem type
 - Data characteristics
 - Desired outcome (interpretability vs. predictive power)

Scenario 1: Bank Loan Approval Model

Problem: A bank needs a model to decide on loan applications. They must be able to explain to a regulator exactly why a loan was denied.

Aspect	Analysis
Problem Type	Classification (approve/deny)
Data	Moderate size; customer financial data
Key Requirement	Interpretability is critical (must explain decisions to regulators)

Aspect	Analysis
Risk	A "black box" model could hide bias
Recommended Model: Decision Tree	
Justification:	
<i>"The highest priority for the bank is interpretability. They must be able to explain exactly why a loan was denied to a customer and to regulators. A Decision Tree is a 'white box' model that provides a clear set of if-then rules for its decisions, making it highly transparent and easy to explain. A more complex model like a neural network, while potentially more accurate, would be a 'black box' and unsuitable for this regulatory requirement."</i>	

Scenario 2: Image Recognition Competition

Problem: A research team wants to enter an image recognition competition where the only goal is to achieve the highest possible accuracy.

Aspect	Analysis
Problem Type	Classification (image classification)
Data	Massive dataset (millions of images)
Key Requirement	Predictive power is the only goal
Risk	None—only accuracy matters
Recommended Model: Convolutional Neural Network (CNN)	
Justification:	
<i>"The only goal is to achieve the highest possible predictive power on a massive image dataset. Explanations do not matter. CNNs are the state-of-the-art architecture specifically designed for image recognition and consistently achieve the highest accuracy on this type of task. The huge dataset size also favours a complex model like a CNN, which can learn intricate patterns that simpler models would miss."</i>	

Scenario 3: Startup Customer Segmentation

Problem: A startup with a small dataset wants to build a simple customer segmentation model.

Aspect	Analysis
Problem Type	Clustering (unsupervised)
Data	Small dataset (200 customers)
Key Requirement	Simple, efficient, no overfitting
Risk	Complex models may overfit

Recommended Model: k-means Clustering

Justification:

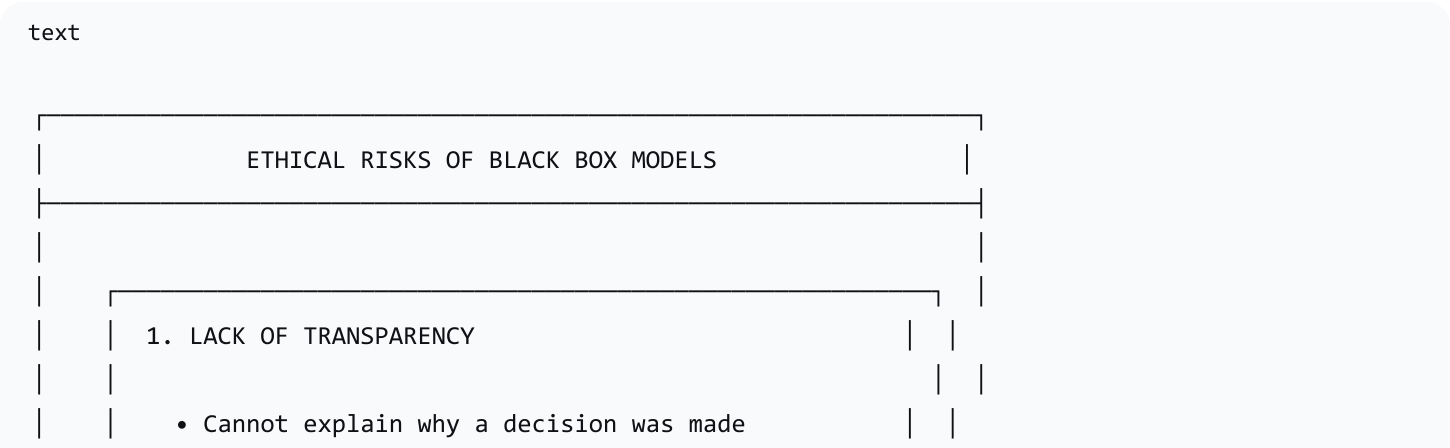
"The problem is to discover unknown groups in the data, not to predict a pre-existing label. This is an unsupervised learning task, making clustering the appropriate choice. Furthermore, with a small dataset of only 200 customers, a simple and efficient algorithm like k-means is a suitable starting point to find basic, actionable segments without the risk of overfitting that a more complex model would have."

6. Ethical Implications of Model Selection

The Ethical Question

"What could happen if the bank used a high-accuracy but 'black box' neural network instead? How does the choice of model connect to the ethical principle of fairness and transparency?"

Ethical Risks of "Black Box" Models



- Cannot prove decisions weren't discriminatory
- Hard to audit for bias

2. REINFORCEMENT OF BIAS

- Learns subtle biases from training data
- Discriminatory patterns go undetected
- Bias is amplified over time

3. NO RECOURSE FOR THE CUSTOMER

- Cannot appeal a decision
- Cannot understand what to change
- Power imbalance

Model Choice as an Ethical Decision

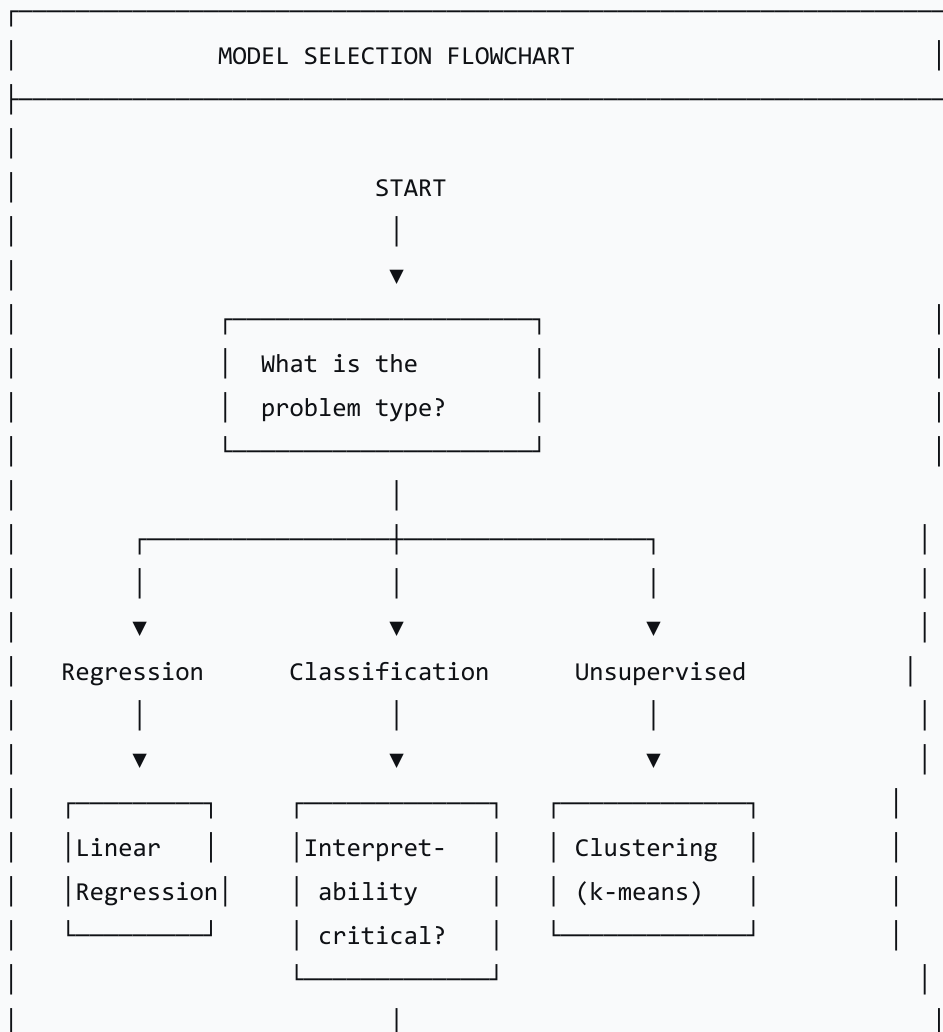
Ethical Principle	Model Selection Connection
Transparency	Choose interpretable models (white box) over black box models
Fairness	Choose models that can be audited for bias
Accountability	Choose models that allow for explanation and appeals
Non-discrimination	Choose models that can detect and correct bias

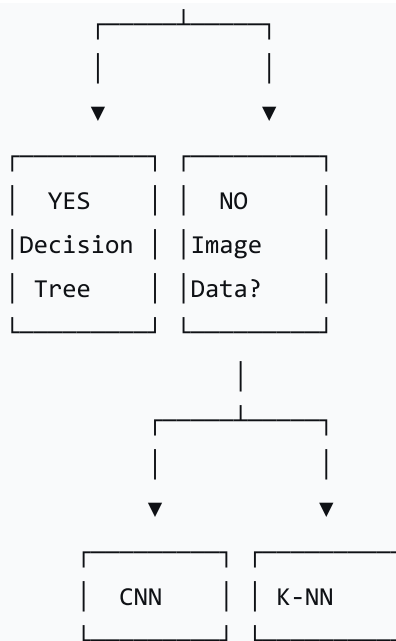
Discussion Points

Question	Purpose
"How does model choice affect fairness?"	Connect model selection to ethics
"What are the risks of using a black box model in healthcare?"	Apply ethical thinking to specific domains
"Who is accountable for a model's decision?"	Discuss accountability
"How can we ensure models are transparent and fair?"	Discuss solutions

7. Model Selection Flowchart

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8. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Recap & Hook	10 min	Recap all models learned; "Mechanic's Toolbox" analogy
Lecture: No Free Lunch	15 min	No Free Lunch theorem; model selection criteria (problem type, data, desired outcome)
Group Activity	25 min	"Chief Data Scientist for a Day" — groups choose models for three scenarios
Discussion & Ethics	10 min	Review activity answers; connect model choice to fairness and transparency

Discussion Questions

Question	Purpose
"Why is there no single best machine learning model?"	Understand No Free Lunch
"What factors should you consider when selecting a model?"	Identify selection criteria
"When would you choose a decision tree over a neural network?"	Apply trade-off reasoning
"What are the ethical risks of using a black box model?"	Connect model selection to ethics

9. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
"Chief Data Scientist" Activity	Assess ability to select and justify models
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the No Free Lunch theorem?
2. Name three factors to consider when selecting a machine learning model.
3. When would you choose a simpler model over a more complex one?
4. Why is model interpretability important in certain applications?

10. Differentiation

Level	Strategy
Support	Provide a reference sheet of models and their characteristics; use the flowchart
Challenge	Explore ethical frameworks; research real-world cases of algorithmic bias

11. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.3.10 & A4.4.1 with examples)
- **Worksheet:** Model selection scenarios and justifications
- **Worksheet Answers**
- **Quizlet** (A4.3.10 vocabulary flashcards)

12. Summary: Key Takeaways

Concept	Key Point
Model Selection	Choosing the right algorithm for the problem
No Free Lunch	No single model is best for every problem
Problem Type	Regression, classification, clustering, reinforcement learning
Data Characteristics	Size, linearity, features, labelled/unlabeled
Interpretability	Can we understand and explain the model's decisions?
Predictive Power	How accurate is the model?
Trade-off	Interpretability vs. predictive power

Concept

Key Point

Ethics

Model choice has ethical implications (fairness, transparency)

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KEY REMINDERS

- ✓ No Free Lunch = no single best model
- ✓ Consider: problem type, data, desired outcome
- ✓ Interpretability vs. predictive power = trade-off
- ✓ Decision Tree = interpretable ("white box")
- ✓ Neural Network = powerful but not interpretable
- ✓ Model choice has ethical implications
- ✓ Choose transparency and fairness when possible

"Selecting the right machine learning model is a balancing act—considering accuracy, interpretability, data characteristics, and the ethical implications of your choice."